No.	Status	Date
LNDE-RPT-01-037-DS-4-RA-101	Issued	2001-03-02

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-101 located within the ATU-100. The scope of work is limited to the thickness data collection for this specific rich amine line as part of the broader survey detailed in LNDE-RPT-01-037.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Basis

All thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 0, the site NDE procedure issued for ATU-100 pre-startup and in-service NDE.

The measurement techniques and equipment calibration were conducted in accordance with the ASTM E797 standard, edition 95.

Line Specifications

The subject line, 4-RA-101, is constructed from A106 Gr.B with a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm.

The design limits for this line are established at 862 kPa(g) and 82 °C. The minimum allowable thickness (t-min) is 3.81 mm.

Assessment Criteria

The retirement thickness for 4-RA-101 is defined as the t-min of 3.81 mm. Any measured value falling below this threshold is subject to an NCR.

Corrosion rates are determined by the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart. Remaining life is calculated as (t - t-min) / rate.

Inspection Interval

The next thickness reading is due at no more than half the remaining life, subject to the class maximum. For this rich amine service, the class maximum for thickness and external visual inspection is 10 / 5 years.

This data sheet supports the 5-year Class 1 survey and rich amine piping assessment documented in LNDE-RPT-01-037.

Thickness readings, measured 2001-03-02

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	1996-03-08	7.12	6.82	0.060	0.060	>50	2006-03-02	
6-SG-101-02	7.11	3.56	1996-03-08	7.05	6.71	0.068	0.068	46.2	2006-03-02	
6-SG-101-03	7.11	3.56	1996-03-08	7.08	6.74	0.068	0.068	46.6	2006-03-02	
6-SG-101-04	7.11	3.56	1996-03-08	7.09	6.73	0.072	0.072	43.9	2006-03-02	
6-SG-101-05	7.11	3.56	1996-03-08	7.14	6.68	0.092	0.092	33.8	2006-03-02	
6-SG-101-06	7.11	3.56	1996-03-08	7.07	6.87	0.040	0.040	>50	2006-03-02	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	1996-03-08	7.14	6.72	0.084	0.084	37.5	2006-03-02	
6-SG-102-02	7.11	3.56	1996-03-08	7.11	6.72	0.078	0.078	40.4	2006-03-02	
6-SG-102-03	7.11	3.56	1996-03-08	7.04	6.69	0.070	0.070	44.6	2006-03-02	
6-SG-102-04	7.11	3.56	1996-03-08	6.99	6.64	0.070	0.070	43.8	2006-03-02	
6-SG-102-05	7.11	3.56	1996-03-08	7.13	6.93	0.040	0.040	>50	2006-03-02	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	1996-03-08	8.52	8.39	0.026	0.026	>50	2011-03-02	
4-RA-101-02	8.56	3.81	1996-03-08	8.54	8.45	0.018	0.018	>50	2011-03-02	
4-RA-101-03	8.56	3.81	1996-03-08	8.61	8.42	0.038	0.038	>50	2011-03-02	
4-RA-101-04	8.56	3.81	1996-03-08	8.63	8.37	0.052	0.052	>50	2011-03-02	
4-RA-101-05	8.56	3.81	1996-03-08	8.60	8.42	0.036	0.036	>50	2011-03-02	
4-RA-101-07	8.56	3.81	1996-03-08	8.60	8.51	0.018	0.018	>50	2011-03-02	
4-RA-101-08	8.56	3.81	1996-03-08	8.68	8.30	0.076	0.076	>50	2011-03-02	
4-RA-101-09	8.56	3.81	1996-03-08	8.57	8.43	0.028	0.028	>50	2011-03-02	
4-RA-101-10	8.56	3.81	1996-03-08	8.67	8.48	0.038	0.038	>50	2011-03-02	
4-RA-101-11	8.56	3.81	1996-03-08	8.68	8.39	0.058	0.058	>50	2011-03-02	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	1996-03-08	8.54	8.48	0.012	0.012	>50	2011-03-02	
4-RA-102-02	8.56	3.81	1996-03-08	8.57	8.50	0.014	0.014	>50	2011-03-02	
4-RA-102-03	8.56	3.81	1996-03-08	8.56	8.37	0.038	0.038	>50	2011-03-02	
4-RA-102-04	8.56	3.81	1996-03-08	8.54	8.35	0.038	0.038	>50	2011-03-02	
4-RA-102-05	8.56	3.81	1996-03-08	8.55	8.36	0.038	0.038	>50	2011-03-02	
4-RA-102-06	8.56	3.81	1996-03-08	8.51	8.38	0.026	0.026	>50	2011-03-02	
4-RA-102-07	8.56	3.81	1996-03-08	8.58	8.49	0.018	0.018	>50	2011-03-02	
4-RA-102-08	8.56	3.81	1996-03-08	8.49	8.36	0.026	0.026	>50	2011-03-02	
4-RA-102-09	8.56	3.81	1996-03-08	8.55	8.35	0.040	0.040	>50	2011-03-02	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	1996-03-08	11.00	10.82	0.036	0.036	>50	2011-03-02	
6-RA-103-02	10.97	4.57	1996-03-08	11.09	10.69	0.080	0.080	>50	2011-03-02	
6-RA-103-03	10.97	4.57	1996-03-08	10.99	10.84	0.030	0.030	>50	2011-03-02	
6-RA-103-04	10.97	4.57	1996-03-08	10.98	10.59	0.078	0.078	>50	2011-03-02	
6-RA-103-05	10.97	4.57	1996-03-08	10.95	10.61	0.068	0.068	>50	2011-03-02	
6-RA-103-06	10.97	4.57	1996-03-08	11.01	10.86	0.030	0.030	>50	2011-03-02	
6-RA-103-07	10.97	4.57	1996-03-08	10.99	10.79	0.040	0.040	>50	2011-03-02	
6-RA-103-08	10.97	4.57	1996-03-08	10.97	10.83	0.028	0.028	>50	2011-03-02	
6-RA-103-09	10.97	4.57	1996-03-08	11.01	10.85	0.032	0.032	>50	2011-03-02	
6-RA-103-10	10.97	4.57	1996-03-08	11.02	10.78	0.048	0.048	>50	2011-03-02	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	1996-03-08	7.59	7.26	0.066	0.066	>50	2006-03-02	
3-RV-101-02	7.62	3.56	1996-03-08	7.57	7.31	0.052	0.052	>50	2006-03-02	
3-RV-101-03	7.62	3.56	1996-03-08	7.52	7.15	0.074	0.074	48.3	2006-03-02	
3-RV-101-04	7.62	3.56	1996-03-08	7.59	7.18	0.082	0.082	44.0	2006-03-02	
3-RV-101-05	7.62	3.56	1996-03-08	7.67	7.25	0.084	0.084	43.8	2006-03-02	
3-RV-101-06	7.62	3.56	1996-03-08	7.59	7.20	0.078	0.078	46.5	2006-03-02	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	1996-03-08	5.53	5.29	0.048	0.048	>50	2006-03-02	
2-RV-102-02	5.54	2.79	1996-03-08	5.59	5.12	0.094	0.094	24.7	2006-03-02	
2-RV-102-03	5.54	2.79	1996-03-08	5.48	5.24	0.048	0.048	>50	2006-03-02	
2-RV-102-04	5.54	2.79	1996-03-08	5.58	5.33	0.050	0.050	>50	2006-03-02	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	1996-03-08	8.51	8.40	0.022	0.022	>50	2011-03-02	
4-RA-104-02	8.56	3.81	1996-03-08	8.55	8.41	0.028	0.028	>50	2011-03-02	
4-RA-104-03	8.56	3.81	1996-03-08	8.54	8.46	0.016	0.016	>50	2011-03-02	
4-RA-104-04	8.56	3.81	1996-03-08	8.59	8.35	0.048	0.048	>50	2011-03-02	
4-RA-104-05	8.56	3.81	1996-03-08	8.60	8.29	0.062	0.062	>50	2011-03-02	
4-RA-104-06	8.56	3.81	1996-03-08	8.59	8.39	0.040	0.040	>50	2011-03-02	
4-RA-104-07	8.56	3.81	1996-03-08	8.59	8.29	0.060	0.060	>50	2011-03-02	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	2001-03-02	_____